

**MBA
USP
ESALQ**

**DATA
ENGINEERING I**

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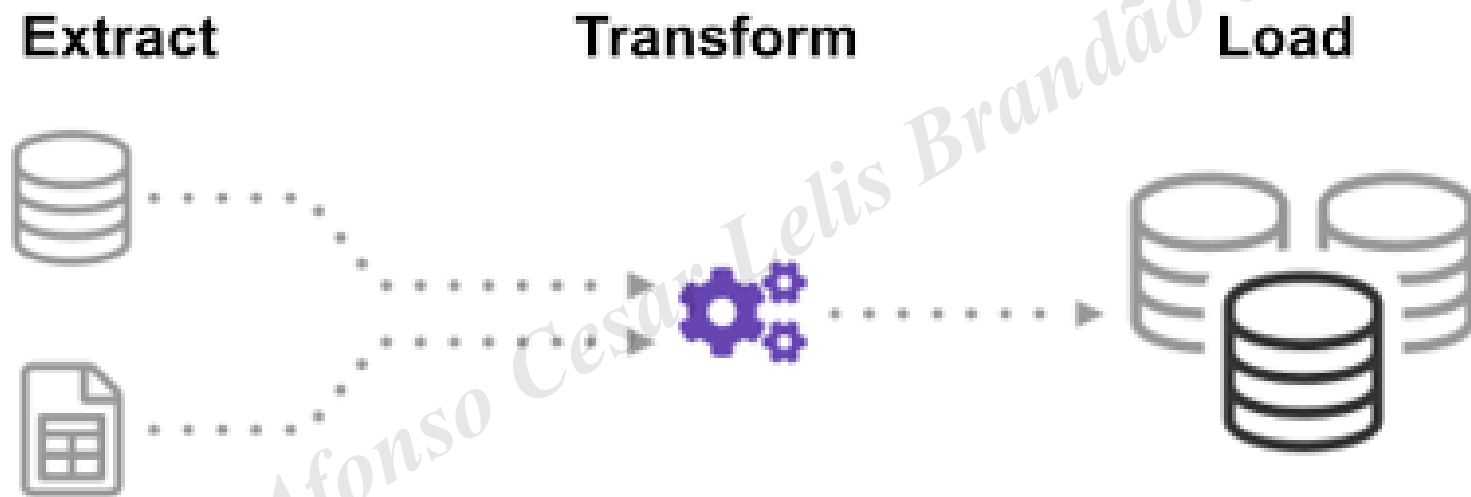
Introduction

Why to study
Data Engineering?

Data Engineering
x Data Science.

What is Data
Engineering?

ETL



Requirements of large companies

A professional of data science must:

- Know about the structures of databases.
- Know how a process of the ETL works.
- Understand about modeling of databases.
- Understand how the use of data in the production works.
- **Know SQL.**

Our objective

Introduction
to the data
structure.

Relational
database.

SQL.

ERD Model –
construction
and
interpretation.

The model and
the relational
algebra.

Data

- Data x Information.
- What is a database?

It is a data collection, which describes, typically, the activities and relationships of one or more organizations.

Example: MBA USP.

DBMS

Database Management Systems:

Software planned to help the maintenance, organization, and collection of existing data in a database.

Example: MySQL.

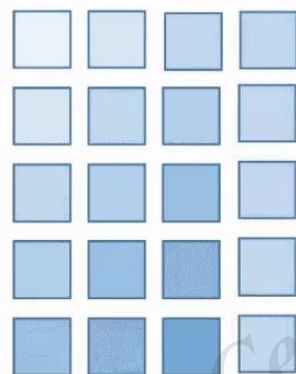
Data Structures

The data we can use are divided into:

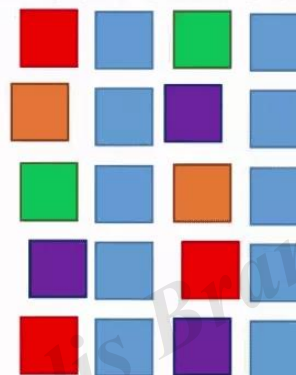
- Structured Data.
- Semi-structured Data.
- Unstructured data.

Structured, Unstructured and Semi-Structured

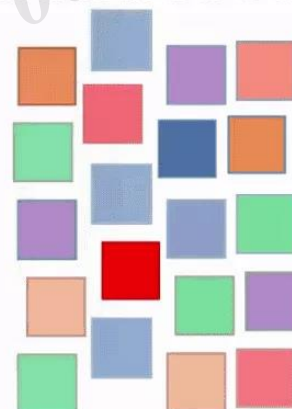
Structured Data



Semi-Structured Data



Unstructured Data



Source: <https://www.astera.com/pt/tipo/blog/dados-semiestruturados-e-n%C3%A3o-estruturados-estruturados/>

Structured the data that have well-defined formats, such as those extracted from spreadsheets or relate databases in SQL.

Semi-structured – Similar to structured data, but not obedient in the totality regarding the form. In this line are the records of languages based on HTML and XML.

Unstructured or NoSQL - do not have a specific format, these are data collected in their original form, such as a text, a video, an email fragment, a system log or a simple photo.

Structured Data

CPF	Name	Grade
x	Zé das couves	10
y	Maria das desgraças	2
h	Silvio Santos	5

Semi-structured data

```
[
  {
    "CPF": "x",
    "Name": "Zé das couves",
    "Grade": "10",
    "Telephone": "It's none of your business"
  },
  {
    "CPF": "y",
    "Name": "Maria das desgraças",
    "Grade": "2"
  },
  {
    "CPF": "h",
    "Name": "Silvio Santos",
    "Grade": "5",
    "Income": "Very high"
  }
]
```

Unstructured Data



Relational DBMS

Our focus will be on Relational DBMS

Advantages in the use of a DBMS:

- Independence.
- Efficiency.
- Integrity and security.
- Simplified data management.
- Access control.

Data model

- Data "stored" in the database according to the model. The DBMS will allow us to look at this model and make consultations according to the pre-established logic.
- The description of the data in terms of models is called **SCHEMA**. As example below:

Students (CPF: string, Name: string, Grade: Integer)

Type of data

- The types of data are classified in different categories and allow N formats. We will present only the most common ones.
- Integer Example: 1, 2, etc.
- Float. Example: 0.10, 10.25, etc.
- String. Example: "Good morning", "my name is", etc.
- Date. Example: 2021-01-01.
- VARCHAR and CHAR case.

Data model

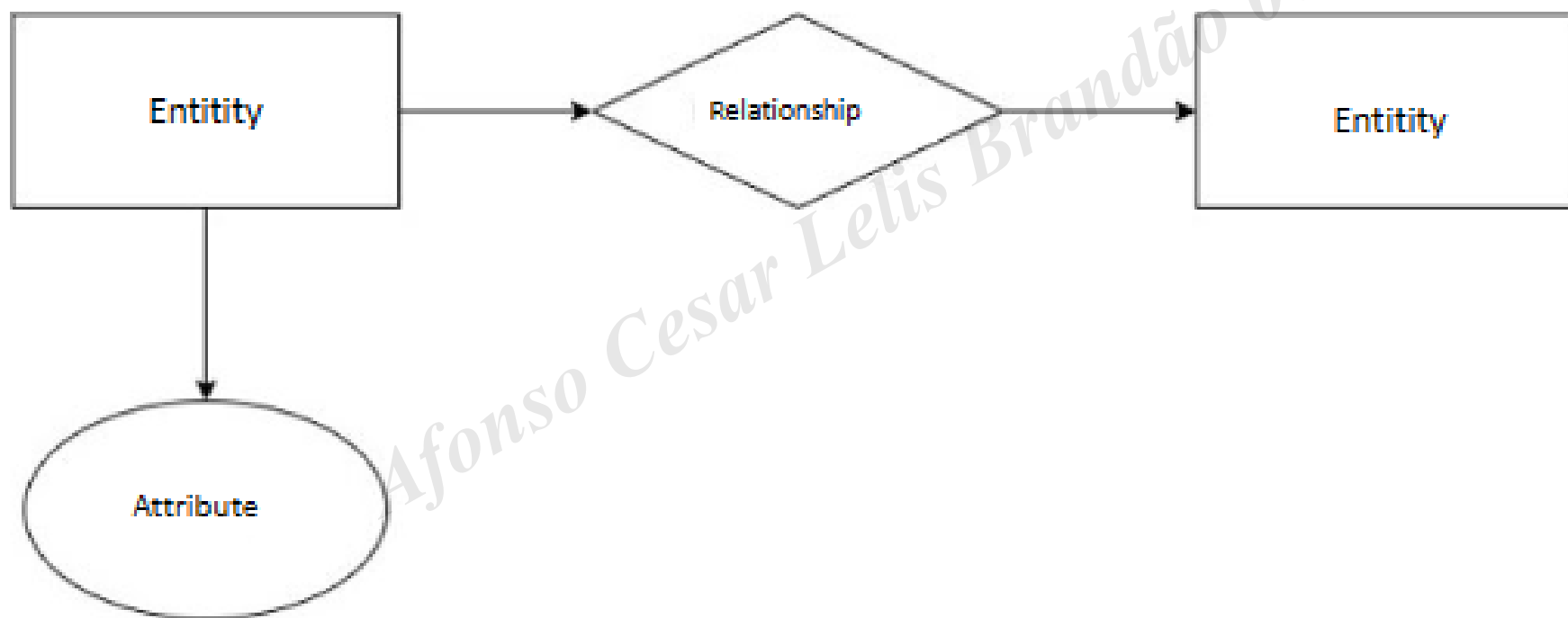
Students (CPF: string, Name: string, Grade: Integer)

- This indicates to us that it is a table with three fields.
- Relational model implies that each register is unique.
- Integrity Constraints!

Level of Abstraction

Conceptual model

- The highest level.
- Closer to the reality of the business.
- It describes the relationships between the entities present in a database.



ERD Definitions

- **Entity:** Something that can be defined and can have data stored about it - such as a person, an object, concept or event. Think on entities as nouns. Examples: a customer, student, car or product.

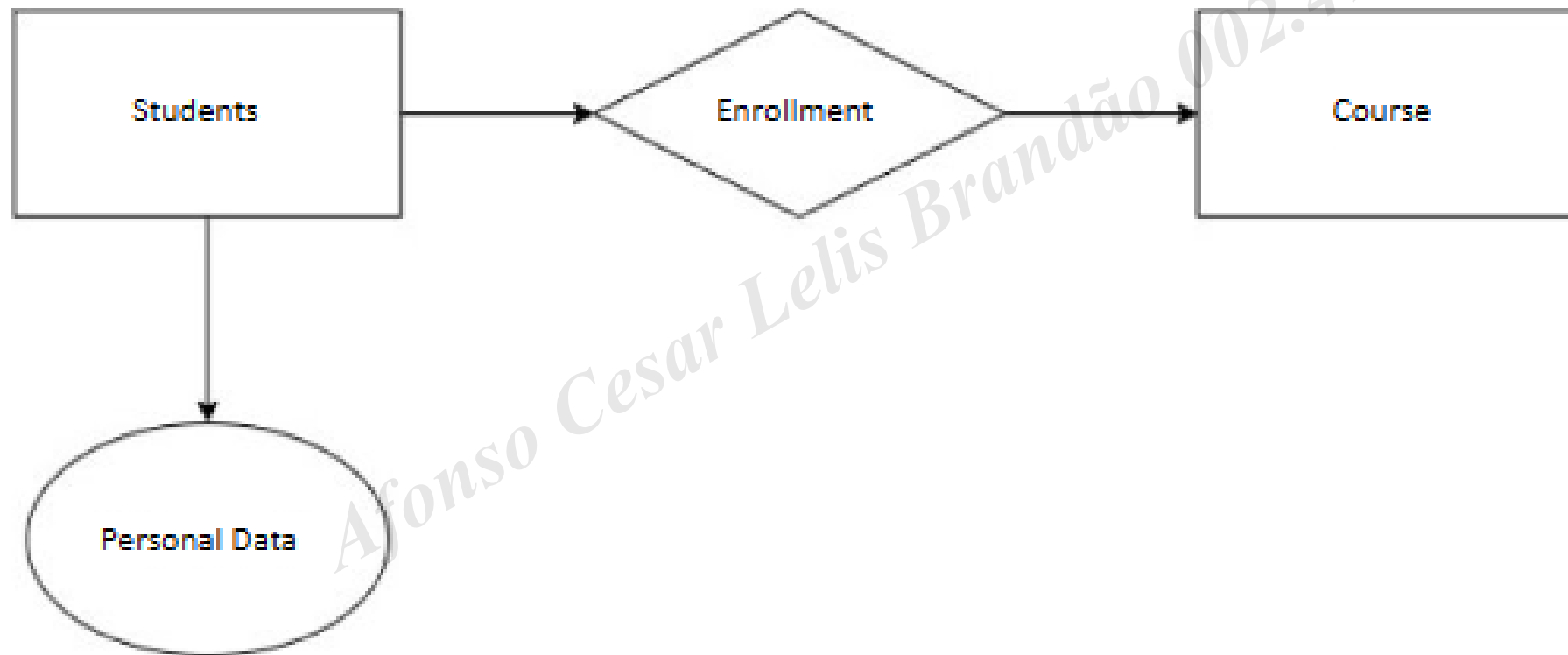
ERD Definitions

- Relationship: How entities act upon each other or are associated with each other. Think on relationships as verbs. For example, the student can sign up in a course. The two entities would be the student and the course, and the relationship described is the act of signing up, so, connecting the two entities.

ERD Definitions

- Attribute: The property or characteristic of an entity, often represented by an oval or circle.

ERD Example

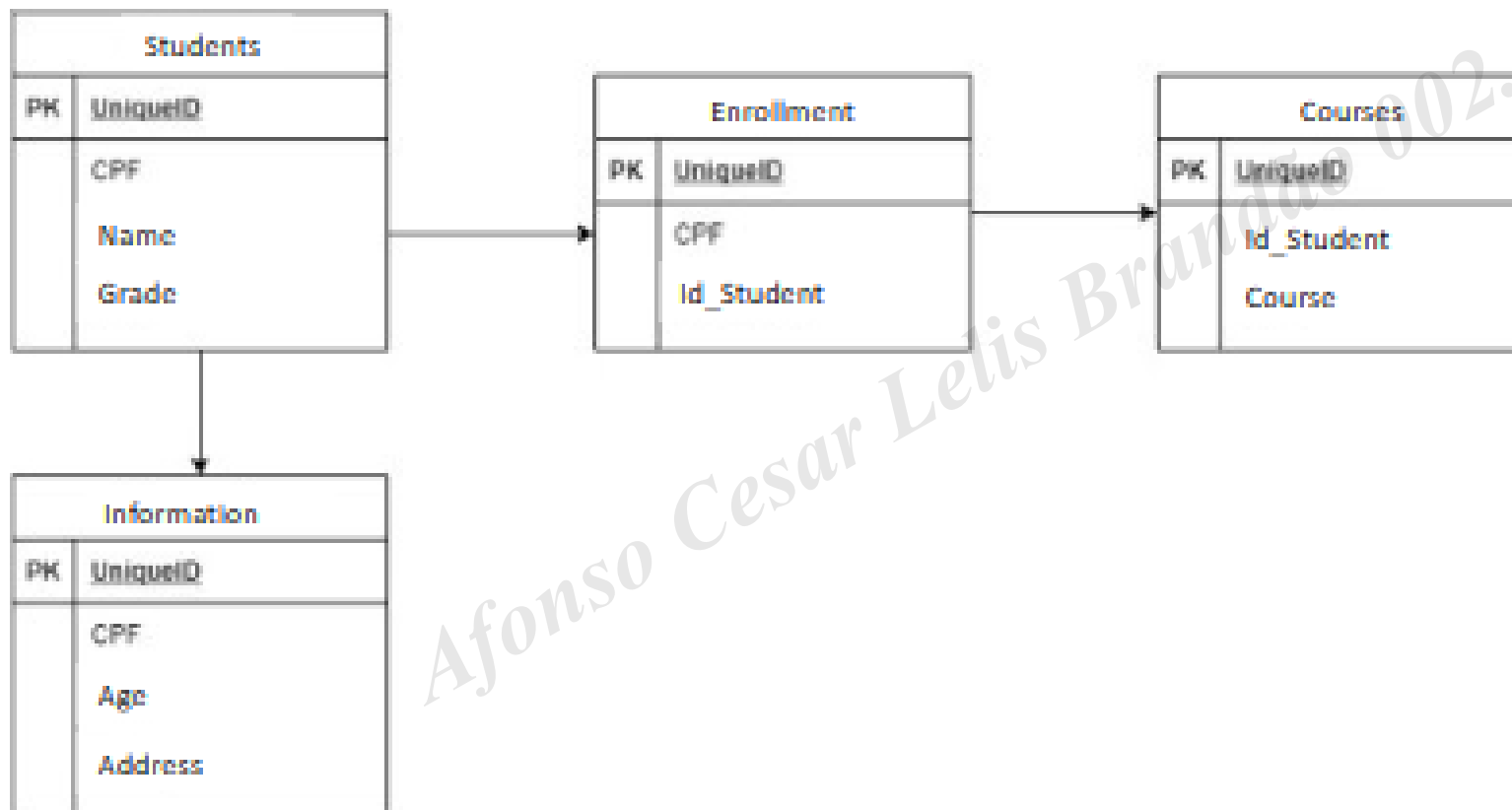


Level of Abstraction

Logical Model

- As the data will be arranged effectively in tables in the database.
- It takes into account limitations of the bank and DBMS.
- Defines the primary and foreigner keys, and integrity constraints.

Logical Schema



Level of Abstraction

Physical Model

- Implementation How to insert the data and create tables and all the schema.
- The lowest level.
- How the data will be stored.
- Restoration methods, backup.

Query

Given the existence of a database, we may ask:

- How many students are enrolled in a course?
- How many courses are active?
- What is the average age of the students?
- What is the average age of students in a certain course?

SQL enters.

SQL

- DML – data manipulation language.
- Universally accepted.
- Proper to use relational algebra

Introduction to SQL

Afonso Cesar Lelis Brandão 002.499.082-55

SQL

- Structured Query Language.
- Origin – IBM.
- We do not need the way to get in the result – we define the result.
- Declarative language.

Important Aspects

DML – data manipulation.

DDL – data definition.

Remote Database Access

Transactions management.

Safety.

Basic form of a query

SELECT [*DISTINCT*] **list of selection**

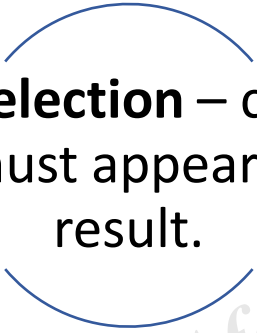
FROM **list of origin**

WHERE **qualification**

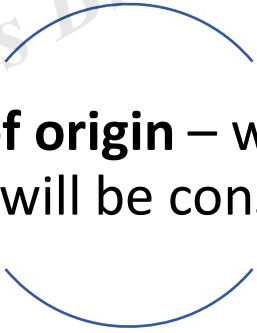
Table: *Students*

CPF	Name	Grade
x	Zé das couves	10
y	Maria das desgraças	2
h	Silvio Santos	5

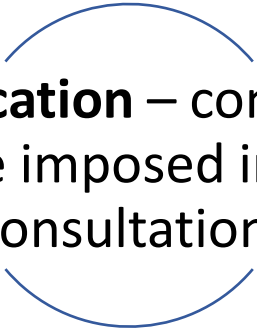
Components



list of selection – columns that must appear in the result.



list of origin – which tables will be consulted.



Qualification – conditions to be imposed in the consultation.

Example 1

- How to obtain a table with CPF and grades?

SELECT CPF, Grade

FROM Students

Observations

- The field's name has to be accurate.
- SQL is case insensitive.
- Separate the name from the columns by commas.

List of origin and Alias

- Alias is the "nickname". It is very used in SQL.
- You can use it to facilitate the understanding of your query.

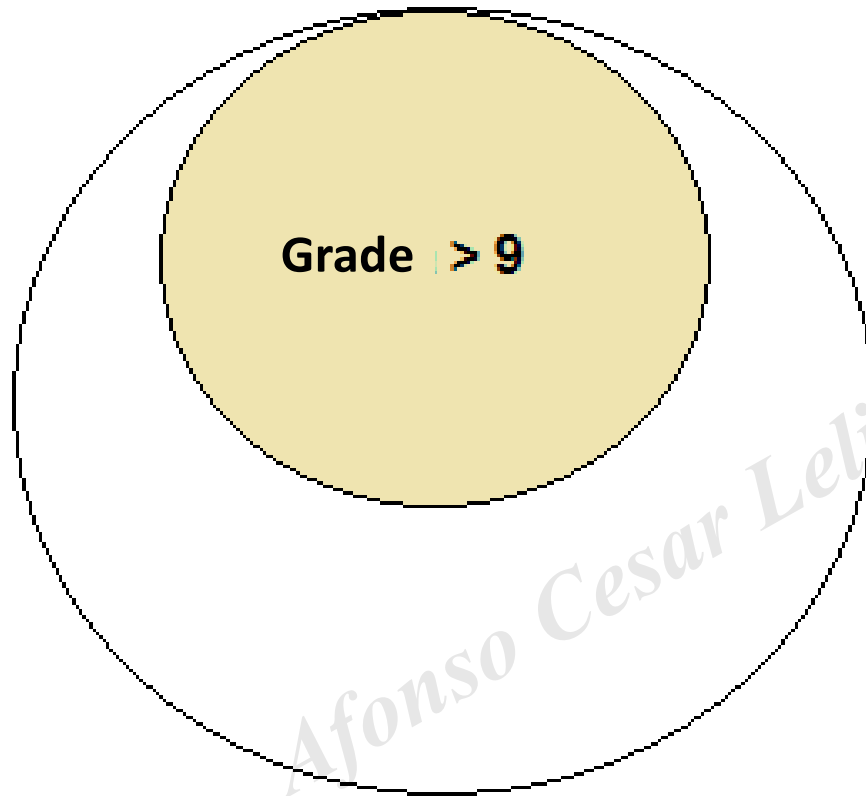
SELECT A.CPF, A.Grade

FROM Students A

Qualification

- The qualification are the "where" clauses.
- These are a Boolean combination of conditions in the form of expressions.
- In terms of algebra, they are definitions of subsets.

Students



Comparison Operators

Operator	Meaning
=	Equal to
> (Maior que)	Greater than
< (Menor que)	Less than
>= (Maior ou igual a)	Greater than or equal to
<= (Menor ou igual a)	Less than or equal to
<> (Diferente de)	It's different to

Example 2

- How to obtain a table with CPF and grades greater than 9?

```
SELECT CPF, Grade  
FROM Students  
WHERE Grade > 9
```

More than a clause

- In this case, we need to define how the relationship is between the clauses.
- Suppose that we have 2 conditions: condition-1 and condition-2.
- AND => the two conditions has to be true at the same time.
- OR => one of the two has to be true.

Operator	Meaning	Example
AND	and	Condition-1 AND condition-2
OR	or	Condition-1 OR condition-2

Example 3

CPF	NAME	GRADE	AGE
XX	JOÃO	10	20
YY	PEDRO	7	30

- How to obtain all records with grades greater than 6 **AND** age greater than 25?

SELECT *

FROM Students

WHERE Grade > 6 AND Age > 25

Example 4

CPF	NAME	GRADE	AGE
XX	JOÃO	10	20
YY	PEDRO	7	30

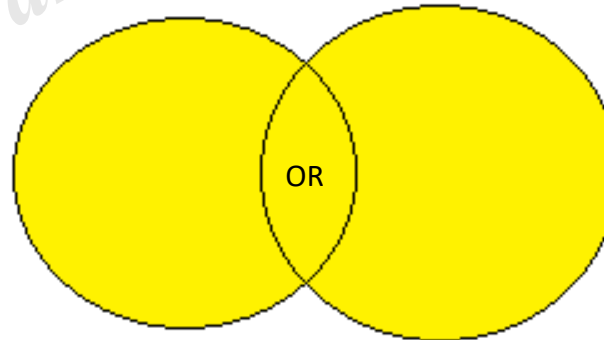
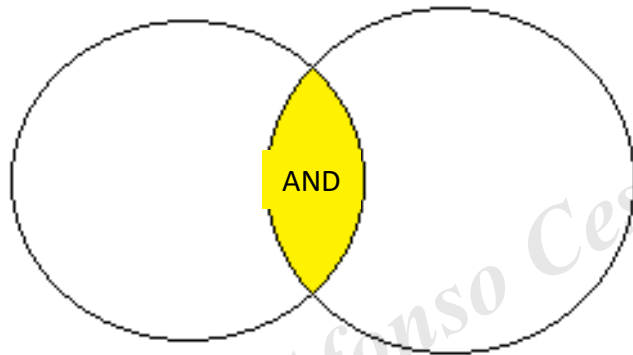
- How to obtain all records with grades greater than 9 **OR** age greater than 25?

SELECT *

FROM Students

WHERE Grade > 9 OR Age > 25

Venn Diagram



Case of repeated lines

PIS	NAME
xxx	Pedro
xxx	Pedro

- Problem in the selection when there are lines with some column of repeated values.
- Use of DISTINCT.

```
SELECT DISTINCT NAME  
FROM Students
```

Aggregation

How to add values per operations.

Data summarization operations:

- Average.
- Minimum.
- Maximum.
- Etc.

COUNT

- It counts the number of records under certain conditions.
- The following logic is presented:

```
SELECT COUNT([Counted Field]), Grouped Fields  
FROM Table  
GROUP BY Grouped Fields
```

Example 5

CPF	NAME	STATE
XXX	JOÃO	SP
YYY	PEDRO	SP
HHH	MARIANA	AL
JJJ	FLAVIA	RJ

SELECT COUNT(CPF)
FROM Students

4

Example 6

CPF	NAME	STATE
XXX	JOÃO	SP
YYY	PEDRO	SP
HHH	MARIANA	AL
JJJ	FLAVIA	RJ

SELECT COUNT(CPF), STATE
FROM Students
GROUP BY STATE

2	SP
1	AL
1	RJ

Example 7

CPF	NAME	STATE
XXX	JOÃO	SP
YYY	PEDRO	SP
HHH	MARIANA	AL
JJJ	FLAVIA	RJ

SELECT COUNT(CPF) THE count, STATE
FROM Students
GROUP BY STATE
ORDER BY count

1	AL
1	RJ
2	SP

SUM

- It sums the number of records under certain conditions.
- The following logic is presented:

```
SELECT SUM([Summed Field]), Grouped Fields  
FROM Table  
GROUP BY Grouped Fields
```

Example 8

CPF	NAME	STATE	AGE
XXX	JOÃO	SP	20
YYY	PEDRO	SP	30
HHH	MARIANA	AL	30
JJJ	FLAVIA	RJ	40

SELECT SUM(AGE)
FROM Students

120

Example 9

CPF	NAME	STATE	AGE
XXX	JOÃO	SP	20
YYY	PEDRO	SP	30
HHH	MARIANA	AL	30
JJJ	FLAVIA	RJ	40

SELECT SUM(AGE), STATE
FROM Students
GROUP BY STATE

50	SP
30	AL
40	RJ

AVERAGE (AVG)

- Calculate the arithmetic average of records under certain conditions.
- The following logic is presented:

```
SELECT AVG([Field]), Grouped Fields  
FROM Table  
GROUP BY Grouped Fields
```


Example 9

CPF	NAME	STATE	AGE
XXX	JOÃO	SP	20
YYY	PEDRO	SP	30
HHH	MARIANA	AL	30
JJJ	FLAVIA	RJ	40

SELECT AVG(AGE), STATE
FROM Students
GROUP BY STATE

25	SP
30	AL
40	RJ

MIN and MAX

- Get the lowest or the highest value of the records under certain conditions.
- The following logic is presented:

SELECT MIN([Field]), Grouped Fields

FROM Table

GROUP BY Grouped Fields

SELECT MAX([Field]), Grouped Fields

FROM Table

GROUP BY Grouped Fields

Example 10

CPF	NAME	STATE	AGE
XXX	JOÃO	SP	20
YYY	PEDRO	SP	30
HHH	MARIANA	AL	30
JJJ	FLAVIA	RJ	40

SELECT MAX(AGE), STATE
FROM Students
GROUP BY STATE

30	SP
30	AL
40	RJ

Example 11

CPF	NAME	STATE	AGE
XXX	JOÃO	SP	20
YYY	PEDRO	SP	30
HHH	MARIANA	AL	30
JJJ	FLAVIA	RJ	40

SELECT MIN(AGE), STATE
FROM Students
GROUP BY STATE

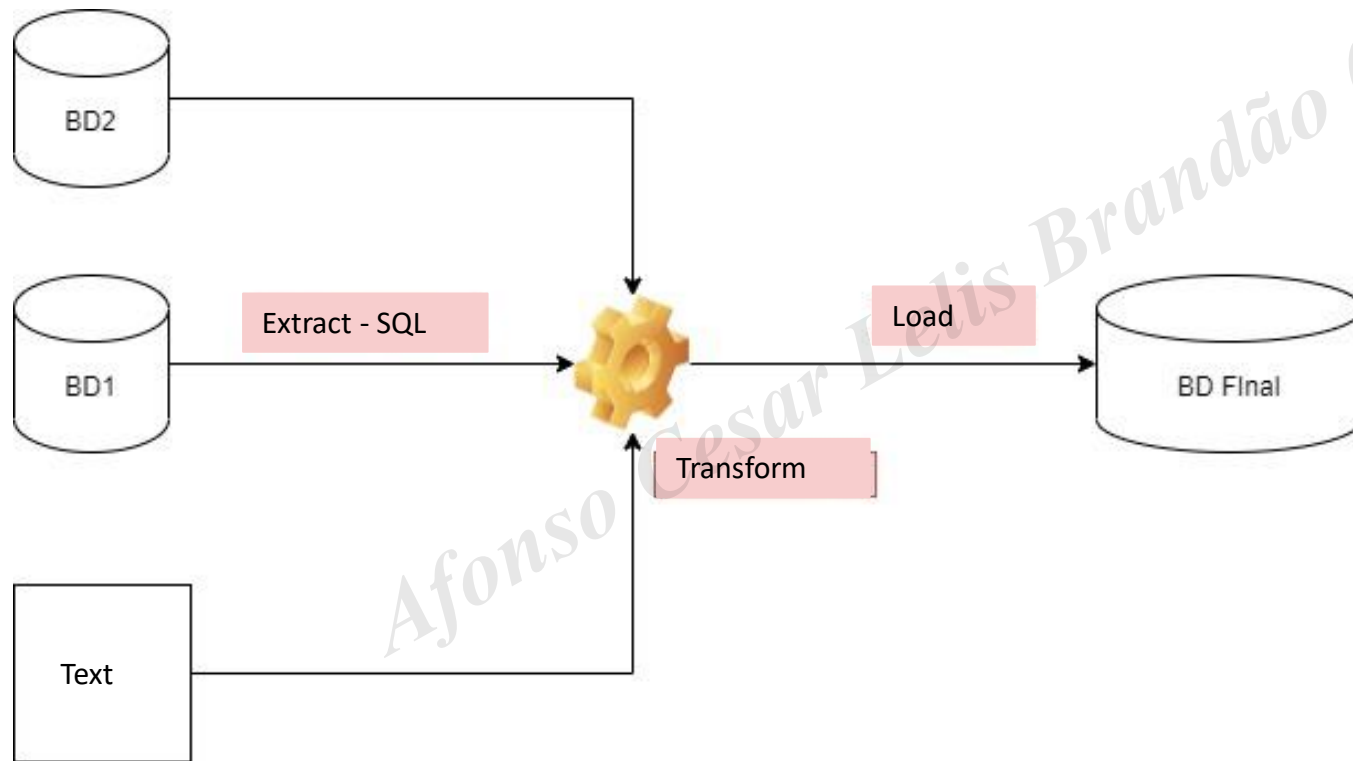
20	SP
30	AL
40	RJ

Introduction to ETL

Extract, Transformation and Load.

The integration of ETL data is a three-step process in which the data is extracted from one or more data sources, converted to the necessary state and loaded into a database or data warehouse in cloud.

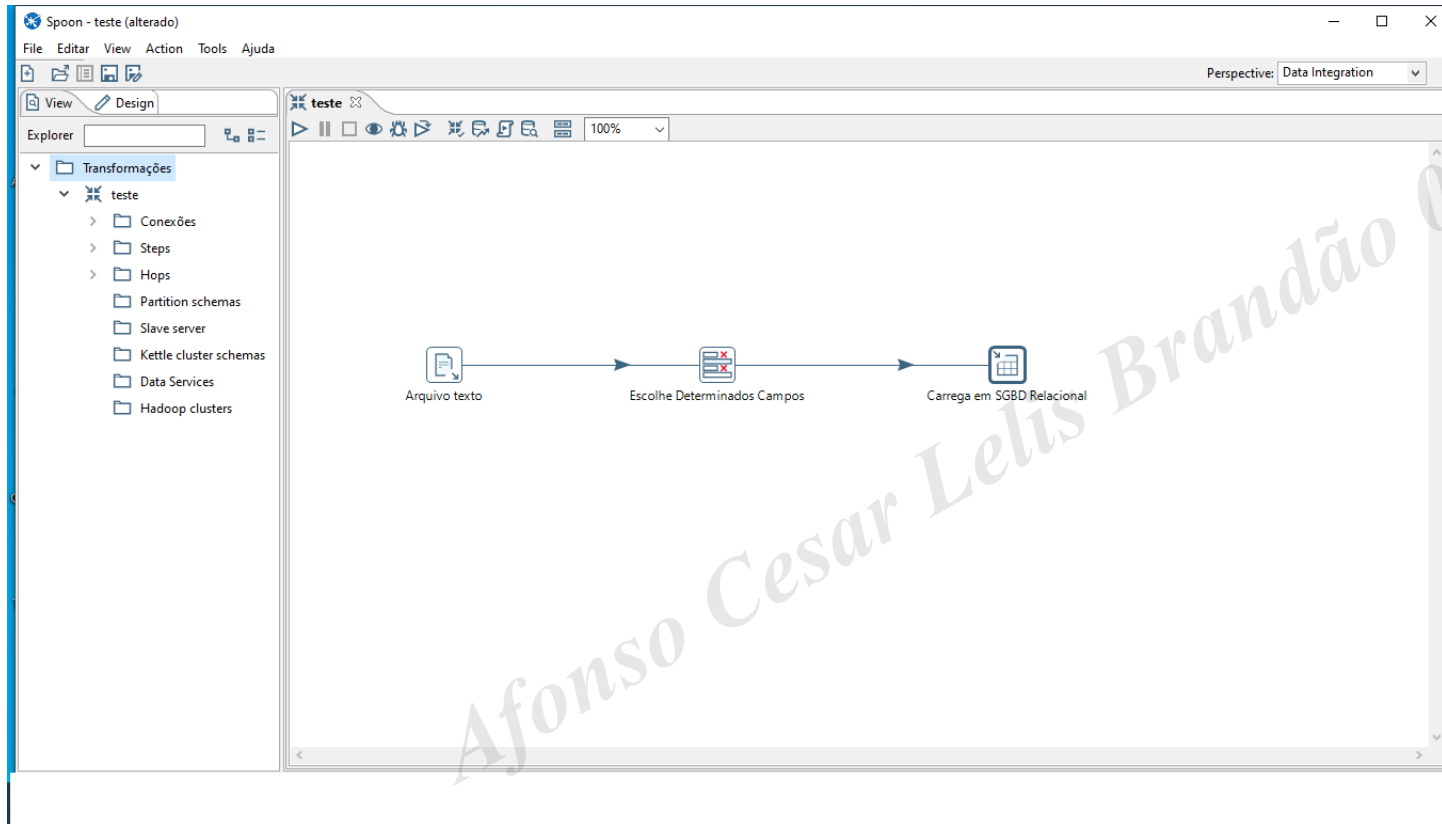
Structure of an ETL



Frameworks for ETL



Example Pentaho



Create Table

- Create a Table with certain fields.
- The following logic is presented:

CREATE TABLE ADDRESS

(
Id_Student INTEGER,
Address Varchar(50)
)

Insert

- Insert certain fields.
- The following logic is presented:

```
INSERT INTO ADDRESS (Id_Student, Address)  
VALUES (2, "RUA PEIXOTO DA SILVA")
```

THANK YOU!

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